

# Mohan Yang

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## EDUCATION

University of California, Los Angeles

Sep 2026 – Jun 2031 (expected)

Ph.D. in Computer Science | Emphasis in Computational Biology

University of California, Irvine

Sep 2022 – Jun 2026

B.S. in Data Science | Minor: Bioinformatics

## PUBLICATIONS

Yang, M.\*, Tong, L.\*, Guo, Z., Tan, Z., Holmes, T.C., Yu, Z., Xu, X. (2026). Evaluating the potential of acupuncture for Alzheimer's disease treatment: A meta-analysis and systematic review of mouse model studies. *Translational Psychiatry*.

Liu, J.\*, Yang, M.\*, Nishida, K. A., Zhang, J. (2026). RNAFlux: A continuous normalizing flow model for RNA velocity. *In Preparation*.

Yang, M., Kanekar, A., Gaona Partida, P.R., Pantula, A., Chen, S., Zuniga, N., Nyembwe, A., Kenney, A. (2026). Early Detection of Polycystic Ovary Syndrome Using the All of Us Research Program. *In Preparation*.

Huynh, K.\*, Yang, M.\*, Rodríguez Ortiz, C., Grieco, S., Thakker, K., Yu, Z., Xu, X. (2026). MoSeq Reveals Behavioral Syllable Shifts after Ketamine in a CORT-Induced Depression Mouse Model. *In Preparation*.

## RESEARCH EXPERIENCE

UCI School of Informatics and Computer Sciences

Irvine, CA

Advisors: Junhao Liu, Prof. Jing Zhang

Dec 2024 – Present

*Generative Model for RNA Velocity*

- Developed a deep generative modeling framework in PyTorch to characterize dynamic cell-state transitions from scRNA-seq data and infer latent cellular progression over continuous time.
- Designed a Neural ODE / continuous normalizing flow decoder to model coupled gene-expression dynamics, improving biological realism in trajectory reconstruction compared with deterministic RNA-velocity assumptions.
- Built reproducible training and evaluation pipelines for high-dimensional single-cell data using Tyro and Slurm, enabling structured experiment configuration, automated testing, and experiment tracking.
- Generated interpretable representations of cellular trajectories that support downstream biological analysis of gene-expression programs and temporal cell-state changes.

UCI School of Medicine & School of Informatics and Computer Sciences

Irvine, CA

Advisors: Prof. Xiangmin Xu, Prof. Zhaoxia Yu

Apr 2024 – Present

*Quantitative Evidence Synthesis: Electroacupuncture in Alzheimer's Disease*

- Methods: Performed random-effects and network meta-analyses (R: *metafor*, *netmeta*); computed SMDs, ran subgroup/heterogeneity and publication-bias assessments, and visualized evidence with forest/funnel plots, network graphs, and Venn diagrams.
- Findings: Acupuncture improved cognition and reduced A $\beta$  deposition, p-tau, and neuroinflammatory biomarkers in AD transgenic mice; Identified the most effective acupoints and found that multi-acupoint protocols do not uniformly outperform single-acupoint treatments.

*ML Tool for Automated Extraction of Scientific Figure Data*

- Built a full-stack ML-powered web application to automatically extract structured data (titles, bar heights, error bars, treatment groups) from journal plots, achieving 95% + accuracy across 200+ test plots.
- Deployed in-lab to streamline data collection for meta-analysis, reducing extraction time from hours to minutes and standardizing dataset generation.

*Behavioral Phenotyping of CORT Response in Mouse Model*

- Analyzed 8 TB of depth-video data using Motion Sequencing to characterize behavioral state changes across treatment progression, and visualized results with transition heatmaps, network plots, and dendrograms.
- Identified significant CORT-induced behavioral changes, including shifts in immobility, locomotion, rearing, and grooming.

## UCI School of Informatics and Computer Sciences

Advisor: Prof. Ana Maria Kenney

### *Early Detection of Polycystic Ovary Syndrome (PCOS)*

- Developed an interpretable longitudinal risk modeling framework using Python and R for early PCOS detection using NIH All of Us EHR data, integrating repeated clinical measurements and visit-level phenotypes to identify risk signals before formal diagnosis.
- Implemented a parametric empirical Bayes approach to model patient-specific clinical trajectories, transforming longitudinal BMI, hyperandrogenism, and testosterone data into visit-level deviation scores for screening.
- Identified early PCOS risk signals up to 2.3 years before diagnosis on average and improved discrimination with multi-feature integration (AUROC 0.73).

Irvine, CA

May 2025 – Present

## UCLA Neurology Department

Advisor: Prof. Jason Hinman

### *Axon Regeneration classification*

- Built an R-based pipeline to process scRNA-seq data that performs quality control, filtering, normalization, variable feature selection, PCA/UMAP using Cell Ranger and Seurat.
- Applied Garnett classifier to categorize cells into regenerating, non-regenerating groups based on gene expression profiles. Quantified regenerative capacity across treatment groups and performed hypothesis testing to assess group differences in regenerative outcomes.

Los Angeles, CA

Sep 2024 – Apr 2025

## UCI School of Informatics and Computer Sciences

Advisor: Prof. Weining Shen

### *GRU-Decay for Early Sepsis Prediction*

- Explored the GRU-Decay model for imputing missing data in multivariate time series, focusing on early sepsis prediction using the PhysioNet Challenge 2019 dataset.
- Implemented the GRU-Decay mechanism using PyTorch, which dynamically adjusts feature imputation based on time gaps, ensuring missing values are weighted between prior observations and empirical means.
- Compared GRU-Decay against Simple GRU and LSTM models, demonstrating superior performance in (AUROC 0.9411) for sepsis prediction.

Irvine, CA

June 2024 – Sep 2024

## PROJECT

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### *Ligand-Based Virtual Screening Model*

- Developed Graph Neural Network models to predict binding affinity for small molecules targeting the Amyloid protein, processed molecular data from the ChEMBL database, converting Canonical SMILES strings into molecular descriptors and graph representations using RDKit and PyTorch Geometric.

## FELLOWSHIP & AWARDS

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- Computing Research Association (CRA) Outstanding Undergrad Research Award 2024-2025
- Donal Bren School of Information & Computer Science Honors Program
- Break Through Tech AI Fellowship at UCLA (2024–25 cohort, ~150 fellows nationwide)
- Undergrad Research Fellowship, Multiple Time Series Integration with Different Resolutions (Summer 2024)
- Undergrad Research Fellowship, Automated extraction tool to Accelerate Meta-Analysis (Summer 2025)
- Undergrad Research Fellowship, Deep Generative Model for Cell Fate Dynamics Modeling (Summer 2025)

## TECHNICAL SKILLS

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**Programming:** Python, C/C++, Java, JavaScript, SQL, R, Stata

**Tools:** AI/ML (PyTorch, TensorFlow, Scikit-learn, Weights & Biases, Slurm, Pandas, NumPy), Bioinfo (Seurat, CellRanger, Severse), Web & Cloud (Docker, AWS, React, Flask, RESTful APIs), Dev Tools: (Git, PyTest)

## LEADERSHIP EXPERIENCE

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### *Learning Assistant for ICS-31 (Introduction to Programming)*

- Led weekly office hours and review sessions; reinforced core Python concepts (data types, control flow, functions) and coached students on course projects. Collaborated with instructors to develop course-related practice activities and exam-prep materials.